

Lab: Sales Performance Analysis of a Retail Store

Problem Statement:

A retail store has provided monthly sales data for three product categories: Electronics, Clothing, and Furniture. The goal is to analyze the trends, compare sales across categories, and identify relationships between the sales of different products using line, bar, and scatter plots.

Objective:

- Load and explore a sales dataset.
- Analyze sales trends over time.
- Compare sales performance across different product categories.
- Visualize sales data using line, bar, and scatter plots.

Requirements:

- Python installed (version 3.x recommended).
- Pandas, NumPy, Matplotlib, and Seaborn installed (`pip install pandas numpy matplotlib seaborn`).
- Jupyter Notebook (optional but recommended for running the lab).
- Retail Sales Dataset from Kaggle: [Retail Sales Dataset](#).

Part 1: Importing Libraries and Loading Data

```
In [18]: # import the required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [20]: # Load the dataset
df = pd.read_csv("Dataset/retail_sales_dataset.csv")

print("Dataset Loaded Successfully")
df.head()
```

Dataset Loaded Successfully

Out[20]:

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
0	1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
1	2	2023-02-27	CUST002	Female	26	Clothing	2	500	1000
2	3	2023-01-13	CUST003	Male	50	Electronics	1	30	30
3	4	2023-05-21	CUST004	Male	37	Clothing	1	500	500
4	5	2023-05-06	CUST005	Male	30	Beauty	2	50	100

Part 2: Data Preprocessing

```
In [21]: # Convert Date column to datetime format
df['Date'] = pd.to_datetime(df['Date'])
df.head()
```

Out[21]:

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
0	1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
1	2	2023-02-27	CUST002	Female	26	Clothing	2	500	1000
2	3	2023-01-13	CUST003	Male	50	Electronics	1	30	30
3	4	2023-05-21	CUST004	Male	37	Clothing	1	500	500
4	5	2023-05-06	CUST005	Male	30	Beauty	2	50	100

```
In [22]: # Aggregate sales data by month and category
df_grouped = df.groupby([df['Date'].dt.to_period('M'), 'Product Category'])['Total Amount'].sum()
df_grouped.index = df_grouped.index.astype(str) # Convert Period index to string

print("Aggregated Monthly Sales Data:")
print(df_grouped.head())
```

```
Aggregated Monthly Sales Data:
Product Category  Beauty  Clothing  Electronics
Date
2023-01          12430.0   13125.0         9895.0
2023-02          14035.0   14560.0        15465.0
2023-03          10545.0   15065.0         3380.0
2023-04          11905.0   13940.0         8025.0
2023-05          12450.0   17455.0        23245.0
```

Part 3: Sales Trends Over Time

```
In [24]: plt.figure(figsize=(12,6))
df_grouped.plot(kind='line', marker='o', figsize=(12,6))
plt.title('Monthly Sales Trends by Product Category')
plt.xlabel('Month')
plt.ylabel('Total Sales')
plt.legend(title='Product Category')
plt.grid(True)
plt.show()
```

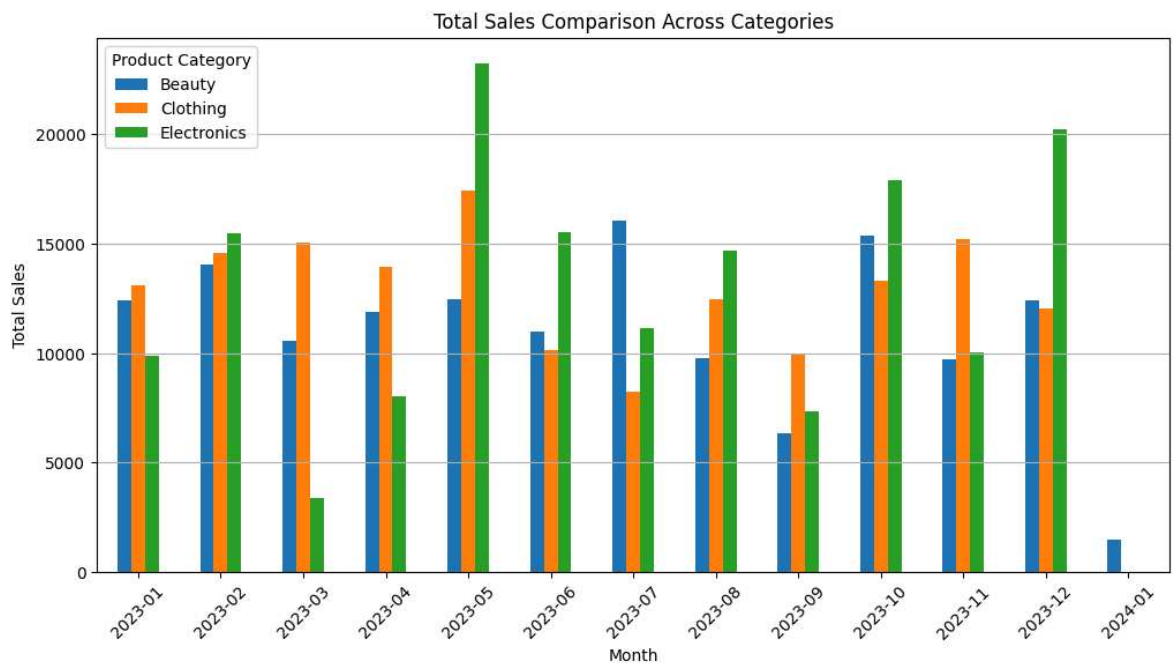
<Figure size 1200x600 with 0 Axes>



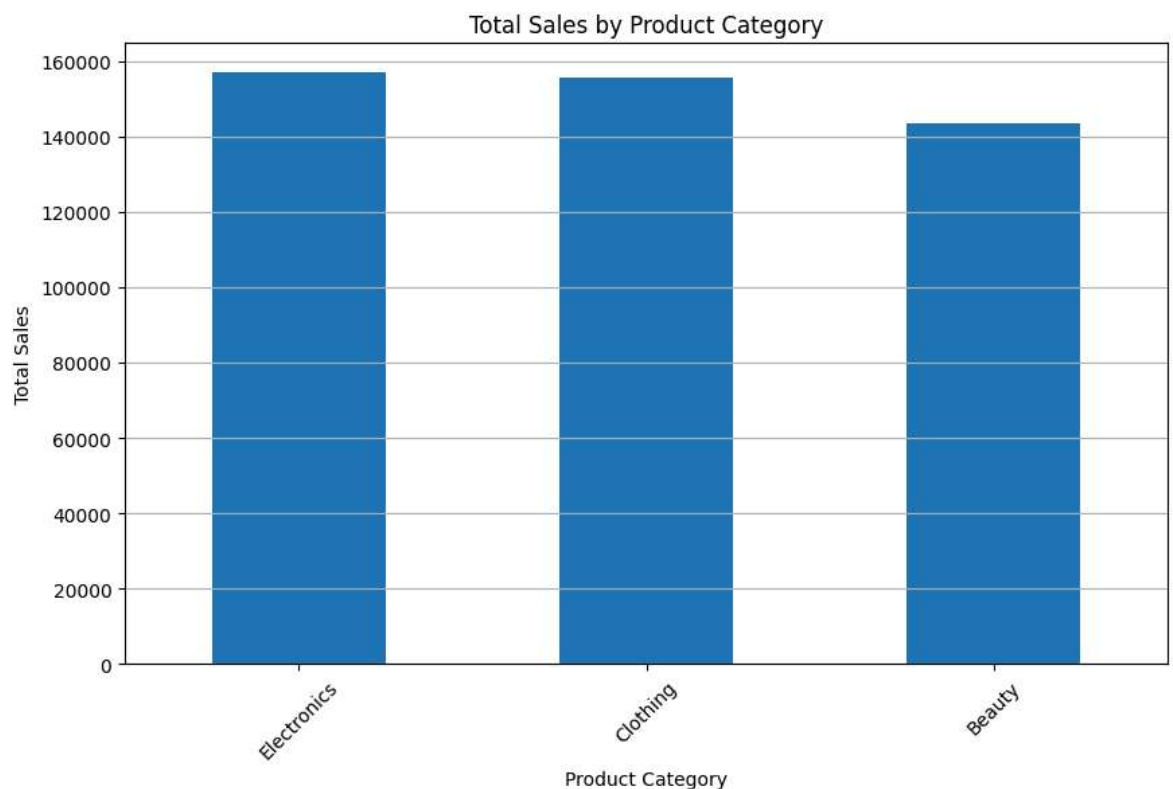
Part 4: Comparing Sales Across Categories

```
In [25]: plt.figure(figsize=(10,6))
df_grouped.plot(kind='bar', figsize=(12,6))
plt.title('Total Sales Comparison Across Categories')
plt.xlabel('Month')
plt.ylabel('Total Sales')
plt.legend(title='Product Category')
plt.xticks(rotation=45)
plt.grid(axis='y')
plt.show()
```

<Figure size 1000x600 with 0 Axes>



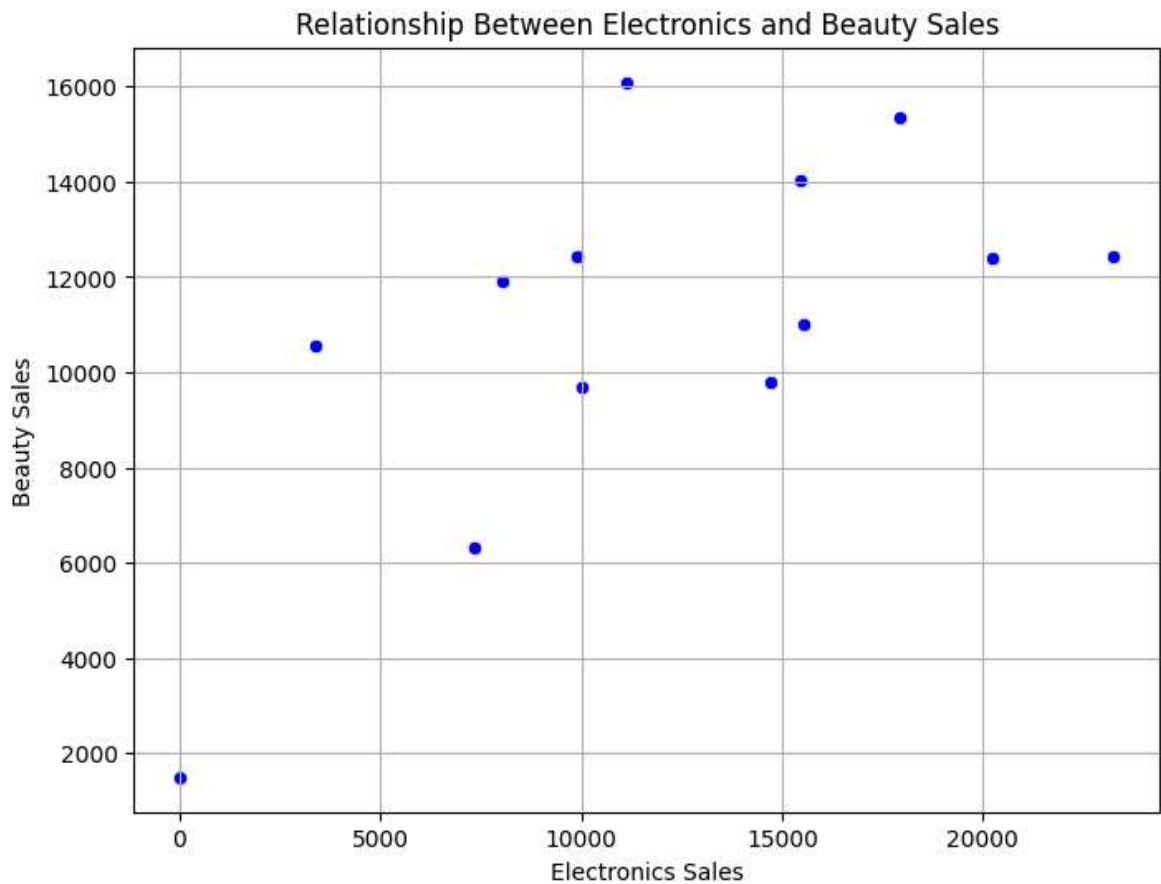
```
In [26]: # overall total sales by product category
df_category_sales = df.groupby('Product Category')['Total Amount'].sum().sort_values(ascending=False)
df_category_sales.plot(kind='bar', figsize=(10,6))
plt.title('Total Sales by Product Category')
plt.xlabel('Product Category')
plt.ylabel('Total Sales')
plt.xticks(rotation=45)
plt.grid(axis='y')
plt.show()
```



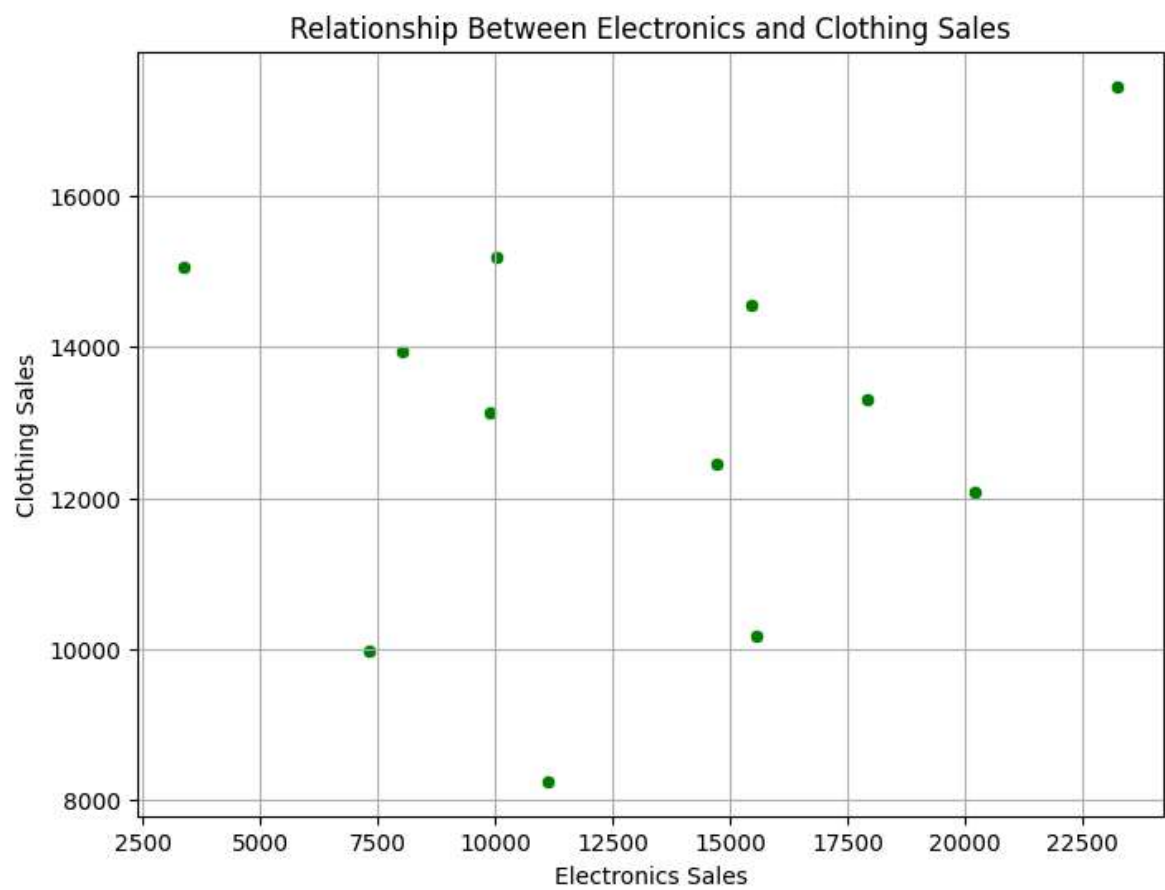
Part 5: Analyzing Relationships Between Product Sales

```
In [28]: plt.figure(figsize=(8,6))
sns.scatterplot(x=df_grouped['Electronics'], y=df_grouped['Beauty'], color='blue')
```

```
plt.title('Relationship Between Electronics and Beauty Sales')
plt.xlabel('Electronics Sales')
plt.ylabel('Beauty Sales')
plt.grid(True)
plt.show()
```



```
In [29]: # analyze the relationship between sales of Electronics and clothin products
plt.figure(figsize=(8,6))
sns.scatterplot(x=df_grouped['Electronics'], y=df_grouped['Clothing'], color='gr')
plt.title('Relationship Between Electronics and Clothing Sales')
plt.xlabel('Electronics Sales')
plt.ylabel('Clothing Sales')
plt.grid(True)
plt.show()
```



In []: